

STAT 652 - Lecture Notes

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1 Exponential Families

Definition (Exponential Density Function).

$$f(x; \lambda) = \lambda^{-1} e^{-\frac{x}{\lambda}}$$

for $\lambda > 0$ **Random Sample:** $X_1, X_2, \dots, X_n$ where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is the sample mean.

Likelihood Function:

$$L(\lambda) = \prod_{i=1}^n f(x_i; \lambda) = \prod_{i=1}^n \lambda^{-1} e^{-\frac{x_i}{\lambda}} = \lambda^{-n} e^{-\frac{\sum_{i=1}^n x_i}{\lambda}} = \lambda^{-n} e^{-\frac{n\bar{x}}{\lambda}}$$

Log-Likelihood Function:

$$l(\lambda) = \log L(\lambda) = \sum_{i=1}^n \log f(x_i; \lambda) = \sum_{i=1}^n \log \lambda^{-1} e^{-\frac{x_i}{\lambda}} = -n \log \lambda - \frac{\bar{x}}{\lambda}$$

MLE:

$$\begin{aligned}\hat{\lambda} &= \bar{x} \\ l'(\lambda) &= -\frac{n}{\lambda} + \frac{n\bar{x}}{\lambda^2} = 0 \\ l''(\lambda) &= \frac{n}{\lambda^2} - \frac{2n\bar{x}}{\lambda^3}\end{aligned}$$

Note that $l''(\hat{\lambda}) = \begin{cases} < 0 & \lambda < 2\bar{x} \quad \text{concave} \\ > 0 & \lambda > 2\bar{x} \quad \text{convex} \end{cases}$ Thus, $\hat{\lambda} = \bar{x}$ is a local maximum. We can prove

that this is a global maximum by showing that $l'(\lambda) > 0$ for $\lambda < \bar{x}$ and $l'(\lambda) < 0$ for $\lambda > \bar{x}$.

Reparamteriation: $\theta = \frac{1}{\lambda} \implies \lambda = \frac{1}{\theta}$

$$f(x; \theta) = \theta e^{-\theta x}$$

Log-Likelihood Function:

$$l(\theta) = \sum_{i=1}^n \log f(x_i; \theta) = \sum_{i=1}^n \log \theta e^{-\theta x_i} = n \log \theta - \theta \sum_{i=1}^n x_i$$

$$l'(\theta) = \frac{n}{\theta} - \sum_{i=1}^n x_i = 0 \implies \hat{\theta} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$$

This is the only critical point and $l''(\theta) = -\frac{n}{\theta^2} < 0$ for $\theta > 0$, so this is a global maximum. And thus $\hat{\lambda} = \frac{1}{\hat{\theta}} = \bar{x}$. is the unique MLE of λ .

Remark (MLE reparametrization). The MLE is invariant under one-to-one reparametrization.

Definition (Normal Density Function).

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for $\mu \in \mathbb{R}$ and $\sigma^2 > 0$.

Random Sample: $X_1, X_2, \dots, X_n$

Likelihood Function:

$$L(\mu, \sigma^2) = \prod_{i=1}^n f(x_i; \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}} = (2\pi\sigma^2)^{-\frac{n}{2}} e^{-\frac{\sum_{i=1}^n (x_i-\mu)^2}{2\sigma^2}}$$

Log-Likelihood Function:

$$l(\mu, \sigma^2) = \log L(\mu, \sigma^2) = \sum_{i=1}^n \log f(x_i; \mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}$$

MLE:

$$\begin{aligned} \hat{\mu} &= \bar{x} \\ \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \\ l_{\mu}(\mu, \sigma^2) &= \frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^2} = 0 \implies \hat{\mu} = \bar{x} \\ l_{\sigma^2}(\mu, \sigma^2) &= -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^4} = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \end{aligned}$$

These are the unique critical points because:

$$\begin{aligned} l_{\mu\mu}(\mu, \sigma^2) &= -\frac{n}{\sigma^2} < 0 \\ l_{\sigma^2\sigma^2}(\mu, \sigma^2) &= \frac{n}{2\sigma^4} - \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^6} \\ l_{\mu\sigma^2}(\mu, \sigma^2) &= -\frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^4} \\ l_{\sigma^2\mu}(\mu, \sigma^2) &= -\frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^4} \end{aligned}$$

The Hessian matrix is:

$$H = \begin{bmatrix} l_{\mu\mu}(\mu, \sigma^2) & l_{\mu\sigma^2}(\mu, \sigma^2) \\ l_{\sigma^2\mu}(\mu, \sigma^2) & l_{\sigma^2\sigma^2}(\mu, \sigma^2) \end{bmatrix} = \begin{bmatrix} -\frac{n}{\sigma^2} & -\frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^4} \\ -\frac{\sum_{i=1}^n (x_i - \mu)}{\sigma^4} & \frac{n}{2\sigma^4} - \frac{\sum_{i=1}^n (x_i - \mu)^2}{\sigma^6} \end{bmatrix}$$

And the determinant of the Hessian is $\neq 0$.

And at $H(\hat{\mu}, \hat{\sigma}^2)$, we have:

$$H(\hat{\mu}, \hat{\sigma}^2) = \begin{bmatrix} -\frac{n}{\hat{\sigma}^2} & 0 \\ 0 & -\frac{n}{2\hat{\sigma}^4} \end{bmatrix} < 0$$

Thus $\bar{x}$ and $\hat{\sigma}^2$ are the unique MLEs of μ and σ^2 respectively.

Definition (Exponential Family). A one-paramter exponential family $\mathcal{P} = \{P_\theta | \theta \in \Theta\}$, P_θ prob measures on $\mathbb{R}^m$, $\Theta \in \mathbb{R}^p$ is an identifiable family of distributions with density or probability mass function of the form

$$f(x; \theta) = \exp(c(\theta)T(x) + d(\theta) + S(x)) I_{\mathcal{A}}(x)$$

Where $c : \Theta \rightarrow \mathbb{R}$ $\mathcal{A} : \Theta \rightarrow \mathbb{R}$ $T : \mathbb{R}^m \rightarrow \mathbb{R}$ $S : \mathbb{R}^m \rightarrow \mathbb{R}$ and $d : \Theta \rightarrow \mathbb{R}$ are known functions. and $\mathcal{A} \subset \mathbb{R}^m$ not dependent of θ aka Regular

Remark. Density = Radon-Nikodim Derivative of P_θ with respect to a σ -finite measure μ on $\mathbb{R}^m$.

Example. $X \sim \text{Exp}(\lambda)$ with $f(x; \lambda) = \lambda^{-1} e^{-\frac{x}{\lambda}}$ for $x > 0$ and $\lambda > 0$.

This is an exponential family with $c(\lambda) = -\frac{1}{\lambda}$, $T(x) = -x$, $S(x) = 0$ and $d(\lambda) = -\log \lambda$ and $\mathcal{A} = (0, \infty)$. Note that this is not a unique representation of the exponential family.

Example. $X \sim \text{Binom}(n, \theta)$ with n known and $\theta \in (0, 1)$ and $f(x; \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}$ for $x = 0, 1, \dots, n$.

This can be rewritten as $f(x; \theta) = \exp\left(x \log \frac{\theta}{1-\theta} + \log \binom{n}{x} + n \log(1 - \theta)\right)$ for $x = 0, 1, \dots, n$. This is an exponential family with $c(\theta) = \log \frac{\theta}{1-\theta}$, $T(x) = x$, $S(x) = \log \binom{n}{x}$ and $d(\theta) = n \log(1 - \theta)$ and $\mathcal{A} = \{0, 1, \dots, n\}$. Notice that the parametrization of $c(\theta)$ is the logit function.

Example. $X \sim \text{Normal}(\mu, \sigma^2)$ with $f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ for $x \in \mathbb{R}$ and $\mu \in \mathbb{R}$ and $\sigma^2 > 0$.

This can be rewritten as $f(x; \mu, \sigma^2) = \exp\left(-\frac{1}{2\sigma^2}x^2 + \frac{\mu}{\sigma^2}x - \frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right)$ for $x \in \mathbb{R}$.

For a fixed σ^2 , this is an exponential family in μ with $c(\mu) = \frac{\mu}{\sigma^2}$, $T(x) = x$, $S(x) = -\frac{1}{2\sigma^2}x^2$, $d(\mu) = -\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$ and $\mathcal{A} = \mathbb{R}$.

For a fixed μ , this is an exponential family in σ^2 with $c(\sigma^2) = -\frac{1}{2\sigma^2}$, $T(x) = x^2$, $S(x) = \frac{\mu}{\sigma^2}x$, $d(\sigma^2) = -\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$ and $\mathcal{A} = \mathbb{R}$.

Definition (k-dim Exponential Family). A k-dim exponential family $\mathcal{P} = \{P_\theta | \theta \in \Theta\}$, P_θ prob measures on $\mathbb{R}^m$, $\Theta \in \mathbb{R}^p$ is an identifiable family of distributing with density or probability mass function of the form

$$f(x; \theta) = \exp\left(\sum_{j=1}^k c_j(\theta)T_j(x) + d(\theta) + S(x)\right) I_{\mathcal{A}}(x)$$

Where k is the smallest integer for which this representation is possible. Note that θ here is a vector of parameters

Example. $X \sim \text{Normal}(\mu, \sigma^2)$

This is a 2-dim/2-param exponential family with $c_1(\mu, \sigma^2) = \frac{\mu}{\sigma^2}$, $c_2(\mu, \sigma^2) = -\frac{1}{2\sigma^2}$, $T_1(x) = x$, $T_2(x) = x^2$, $S(x) = 0$, $d(\mu, \sigma^2) = -\frac{\mu^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)$ and $\mathcal{A} = \mathbb{R}$.

Remark. P_θ is a exponential family in $\theta = \{\theta_1, \theta_2\} \implies \theta_2$ fixed, its exponential family in θ_1 and vice versa. Reverse implication is not true

Example. $X \sim \text{Multinomial}_m(n, \theta)$ $\theta = (\theta_1, \dots, \theta_m)$, $\sum_m \theta_i = 1$, $\theta_i > 0$ and $f(x; \theta) = \frac{n!}{x_1! \dots x_m!} \prod_{i=1}^m \theta_i^{x_i}$ for $x = (x_1, \dots, x_m)$ with $\sum_{i=1}^m x_i = n$ and $x_i \in \{0, 1, \dots, n\}$.

This can be rewritten as $f(x; \theta) = \exp \left(\sum_{i=1}^m x_i \log \theta_i + \log \frac{n!}{x_1! \dots x_m!} \right)$ for $x = (x_1, \dots, x_m)$ with $\sum_{i=1}^m x_i = n$ and $x_i \in \{0, 1, \dots, n\}$.

Thus

$$c_i(\theta) = \log \theta_i, \quad i = 1, \dots, m-1$$

$$T_i(x) = x_i, \quad i = 1, \dots, m-1$$

$$S(x) = \log \frac{n!}{x_1! \dots x_m!}$$

$$d(\theta) = n \log \left(1 - \sum_{i=1}^{m-1} \theta_i \right)$$

$$\mathcal{A} = \{(x_1, \dots, x_m) | x_i \in \{0, 1, \dots, n\}, \sum_{i=1}^m x_i = n\}$$

Lemma 1. A representation of an exponential family corresponds to k -dim iff

$$1. \sum_{i=1}^k a_j c_j(\theta) = a_0 \quad \forall \theta \in \Theta \text{ iff constants } a_0, a_1, \dots, a_k = 0$$

$$2. \sum_{i=1}^k b_j T_j(x) = b_0 \quad \forall x \in \mathcal{A} \text{ iff constants } b_0, b_1, \dots, b_k = 0$$

This basically means that the $1, c_j(\theta)$ are linearly independent and the functions $1, T_j(x)$ are linearly independent.

Example (Examples of Exponential families). Multinomial $m(\mu, \theta_m)$ Where $\sum_i \theta_i = 1$

$$f(x : \theta) = \binom{N}{x_1, \dots, x_n} \prod_i \theta_i^{x_i}$$

We can write this in exponential family from by

$$c_i \log \theta_i | \theta_m, \quad i = 1, \dots, m-1$$

$$T_i(x) = x_i, \quad i = 1, \dots, m-1$$

$$S(x) = \log \binom{N}{x_1, \dots, x_n}$$

$$d(\theta) = N \log \left(1 - \sum_{i=1}^{m-1} \theta_i \right)$$

$$\mathcal{A} = \{(x_1, \dots, x_m) | x_i \in \{0, 1, \dots, N\}, \sum_{i=1}^m x_i = N\}$$

This is cleary a $k = m - 1$ exponential family

Example (Gamma). The Gamma distribution is a 2-dim exponential family with $c_1(\alpha, \beta) = \alpha - 1$, $c_2(\alpha, \beta) = -\frac{1}{\beta}$, $T_1(x) = \log x$, $T_2(x) = x$, $S(x) = 0$, $d(\alpha, \beta) = -\log \Gamma(\alpha) - \alpha \log \beta$ and $\mathcal{A} = (0, \infty)$.

Definition (Full Rank and Curved Exponential Families). A k-dim exponential family is called full rank if the set $\{c_1(\theta), \dots, c_k(\theta) | \theta \in \Theta\}$ if this contains an open subset of $\mathbb{R}^k$. Otherwise, it is called a curved exponential family.

Example (Multinomial). The multinomial distribution is a full rank exponential family because the set $\{c_1(\theta), \dots, c_{m-1}(\theta) | \theta \in \Theta\}$ contains an open subset of $\mathbb{R}^{m-1}$.

Example (Gamma). The gamma distribution is a curved exponential family because the set $\{c_1(\alpha, \beta), c_2(\alpha, \beta) | \alpha > 0, \beta > 0\}$ does not contain an open subset of $\mathbb{R}^2$.

Remark (Curved Exponential Family). $f(x; \mu) = ce^{-(x-\mu)^4}$ Show that this is a curved exponential family with what k

Proof. We can rewrite this as $f(x; \mu) = \exp(-(x-\mu)^4 + \log c)$ for $x \in \mathbb{R}$ and $\mu \in \mathbb{R}$. This is an exponential family with $c(\mu) = -1$, $T(x) = (x-\mu)^4$, $S(x) = 0$, $d(\mu) = \log c$ and $\mathcal{A} = \mathbb{R}$.

This is a 1-dim exponential family because the set $\{c(\mu) | \mu \in \mathbb{R}\}$ does not contain an open subset of $\mathbb{R}$. $\square$

Remark (Random Sample from an exponential family). Let $X_1, \dots, X_n$ be a random sample from an exponential family with density or probability mass function of the form

$$f(x; \theta) = \exp \left(\sum_{j=1}^k c_j(\theta) T_j(x) + d(\theta) + S(x) \right) I_{\mathcal{A}}(x)$$

Then the joint density or probability mass function of the random sample is

$$f(x_1, \dots, x_n; \theta) = \prod_{i=1}^n f(x_i; \theta) = \exp \left(\sum_{j=1}^k c_j(\theta) \sum_{i=1}^n T_j(x_i) + nd(\theta) + \sum_{i=1}^n S(x_i) \right) I_{\mathcal{A}}(x_1, \dots, x_n)$$

Thus the random sample is also an exponential family with $c_j(\theta)$, $T_j(x)$, $S(x)$, $d(\theta)$ and $\mathcal{A}$

$$c_i(\theta) = c_i(\theta), \quad i = 1, \dots, k$$

$$T_i(x_1, \dots, x_n) = \sum_{i=1}^n T_j(x_i), \quad i = 1, \dots, k$$

$$S(x_1, \dots, x_n) = \sum_{i=1}^n S(x_i)$$

$$d(\theta) = nd(\theta)$$

$$\mathcal{A} = \mathcal{A}^n$$

Theorem 1 (Factorization Theorem). Let $X_1, \dots, X_n$ be a random sample from a distribution with density or probability mass function $f(x; \theta)$. Then $T(X_1, \dots, X_n)$ is a sufficient statistic for θ if and only if there exist functions $g(t; \theta)$ and $h(x_1, \dots, x_n)$ such that

$$f(x_1, \dots, x_n; \theta) = g(T(x_1, \dots, x_n); \theta)h(x_1, \dots, x_n)$$

Theorem 2 (Minimal Sufficient Statistic). Let $X_1, \dots, X_n$ be a random sample from a distribution with density or probability mass function $f(x; \theta)$. Then $T(X_1, \dots, X_n)$ is a minimal sufficient statistic for θ if and only if for any two sample points $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$, the ratio

$$\frac{f(x_1, \dots, x_n; \theta)}{f(y_1, \dots, y_n; \theta)}$$

is constant as a function of θ if and only if $T(x_1, \dots, x_n) = T(y_1, \dots, y_n)$.

Example (Example of a Minimal Sufficient Statistic). Let $X \sim N(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$. Then the minimal sufficient statistic for (μ, σ^2) is (x, x^2) .

Let $X \sim N(0, \sigma^2)$ with $\sigma^2 > 0$. Then the minimal sufficient statistic for σ^2 is $\sum x^2$.

Definition (Natural Reparametrization). A k -dim exponential family $\mathcal{P} = \{P_\theta | \theta \in \Theta\}$, P_θ prob measures on $\mathbb{R}^m$, $\Theta \in \mathbb{R}^p$ is an identifiable family of distributing with density or probability mass function of the form

$$f(x; \theta) = \exp \left(\sum_{j=1}^k c_j(\theta) T_j(x) + d(\theta) + S(x) \right) I_{\mathcal{A}}(x)$$

The natural reparametrization of this exponential family is the k -dim exponential family $\mathcal{P} = \{P_\eta | \eta \in \mathbb{R}^k\}$, P_η prob measures on $\mathbb{R}^m$, $\eta \in \mathbb{R}^k$ with density or probability mass function of the form

$$f(x; \eta) = \exp \left(\sum_{j=1}^k \eta_j T_j(x) + d(\eta) + S(x) \right) I_{\mathcal{A}}(x)$$

where $\eta = (c_1(\theta), \dots, c_k(\theta))$ and $d(\eta) = d(\theta)$.

OMG HE TALKED ABOUT STATISTICS ON MANIFOLDS

Remark. $g(n) = e^{-d_o(n)}$ is convex

Definition (Convex Set). A set $C \subset \mathbb{R}^k$ is convex if for any two points $x, y \in C$, the line segment connecting x and y is contained in C . That is, for any $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)y \in C$.

Definition (Concave Set). A set $C \subset \mathbb{R}^k$ is concave if for any two points $x, y \in C$, the line segment connecting x and y is contained in C is less than or equal to the line segment connecting x and y . That is, for any $\lambda \in [0, 1]$, we have $\lambda x + (1 - \lambda)y \leq C$.

Lemma 2. Suppose $g : \Theta \rightarrow \mathbb{R}$ is a concave d

1. Any local max is a global max
2. There are no local min
3. The set of all global max is either empty or convex
4. If Θ is bounded then a max exists on Θ closure and thus if it is also closed it is in Θ .
5. If $\Theta \in R^k$ then a max exists iff g is coercive ie $g(\theta) \rightarrow -\infty$ as $\|\theta\| \rightarrow \infty$

Remark (Exp Family natural form).

$$f(x; \eta) = \exp \left(\sum_{j=1}^k \eta_j T_j(x) + d_0(\eta) + S(x) \right) I_A(x)$$

Where $\eta = \{\eta = (\eta_1 \dots \eta_k) | \eta = (c_1(\theta), \dots, c_k(\theta)), \theta \in \Theta\}$

Lemma 3. The convex closure of η is a valid parameter space for the natural form of the exponential family.

Example. $X \sim \text{Gamma}(\theta, \theta)$ with $f(x; \theta) = \exp((\theta - 1) \log x - \frac{x}{\theta} - \log \Gamma(\theta) - \theta \log \theta)$ for $x > 0$ and $\theta > 0$.

Where $d(\theta) = -\log \theta^\theta \Gamma(\theta)$

or $f(x; \eta) = \exp(\eta_1 \log x + \eta_2 x + d_0(\eta))$ for $x > 0$ and $\eta = (\theta - 1, -\frac{1}{\theta})$ and $d_0(\eta) = -\log \Gamma(\eta_1 + 1) - (\eta_1 + 1) \log(-\eta_2)$

with $\eta = \left\{ (n_1, n_2) | n_1 > 1, n_2 > \frac{-1}{1+n_1} \right\}$

Lemma 4. Uniqueness of Natural parameterizations under affine transformation k - parameter

n_0 represents a natural parameterization for a exp family iff n_0 is an affine transformation of η .

Corollary. $g(\eta) = e^{d_0(\eta)}$ is convex on η and thus $d_0(\eta)$ is concave on η .

Proof. Since $d_0(\eta)$ is a concave function, its exponential $g(\eta) = e^{d_0(\eta)}$ is also convex. $d_0(\eta)$ is concave because the negative of a convex function is concave. □

Example. $X_1 \dots X_n \sim \text{Gamma}(\alpha, \beta)$

$$l(\alpha, \beta) = \sum_{i=1}^n \log f(x_i; \alpha, \beta) = \sum_{i=1}^n \left((\alpha - 1) \log x_i - \frac{x_i}{\beta} - \log \Gamma(\alpha) - \alpha \log \beta \right)$$

Case 1: β known, α unknown.

$$l(\alpha) = \sum_{i=1}^n ((\alpha - 1) \log x_i - \log \Gamma(\alpha) - \alpha \log \beta)$$

$$l'(\alpha) = \sum_{i=1}^n (\log x_i - \psi(\alpha) - \log \beta) = 0$$

$$l''(\alpha) = \sum_{i=1}^n (-\psi'(\alpha)) < 0$$

Thus $\hat{\alpha}$ is a global maximum.

Case 2: α known, β unknown.

$$l(\beta) = \sum_{i=1}^n \left(-\frac{x_i}{\beta} - \alpha \log \beta \right)$$

$$l'(\beta) = \sum_{i=1}^n \left(\frac{x_i}{\beta^2} - \frac{\alpha}{\beta} \right) = 0$$

$$l''(\beta) = \sum_{i=1}^n \left(-\frac{2x_i}{\beta^3} + \frac{\alpha}{\beta^2} \right) < 0$$

Thus $\hat{\beta}$ is a global maximum.

This is sometimes referred to as the digamma function $\psi(\alpha) = \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}$ and the trigamma function $\psi'(\alpha) = \frac{d}{d\alpha} \psi(\alpha)$.

Case 3: α and β unknown.

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Theorem 3. Log-likelihood for an exp family is strictly concave in $\eta \in \eta$

Proof.

□

Lemma 5 (4.1). The mgf exists for any interior point $\eta \in \bar{\eta}$ with

$$M_X(t) = \exp(d_0(\eta + t) - d_0(\eta))$$

Lemma 6 (4.2). The density/mass of $T(X)$ the natural maximum sufficient statistic is given by $f_t(\tau) = \exp\left(\sum_{j=1}^k \eta_j \tau_j + d_0(\eta) + S_0(\tau)\right) I_{\mathcal{T}}(\tau)$

Theorem 4 (A minimal sufficient statistic from a full rank exp family is complete). Essentially this means that if $T(X)$ is a minimal sufficient statistic for a full rank exponential family, then it is also complete. This implies that any unbiased estimator based on $T(X)$ is unique and cannot be improved upon by any other unbiased estimator.

Theorem 5 (Lehmann-Scheffe Theorem). If $T(X)$ is a complete sufficient statistic for a parameter θ , then any unbiased estimator of a function of θ that is based on $T(X)$ is the unique minimum variance unbiased estimator (UMVUE) of that function.

Theorem 6 (Rao-Blackwell Theorem). *If $T(X)$ is a sufficient statistic for a parameter θ , and $\hat{\theta}$ is an unbiased estimator of a function of θ , then the conditional expectation of $\hat{\theta}$ given $T(X)$, denoted as $E[\hat{\theta}|T(X)]$, is also an unbiased estimator of that function and has variance less than or equal to that of $\hat{\theta}$. This means that $E[\hat{\theta}|T(X)]$ is at least as good as $\hat{\theta}$ in terms of variance, and if $T(X)$ is complete, then it is the unique minimum variance unbiased estimator (UMVUE).*

Proof. the proof of the Rao-Blackwell theorem involves showing that the conditional expectation $E[\hat{\theta}|T(X)]$ is unbiased and has variance less than or equal to that of $\hat{\theta}$. Let $T(X)$ be a sufficient statistic for θ , and let $\hat{\theta}$ be an unbiased estimator of a function of θ . We want to show that $E[\hat{\theta}|T(X)]$ is also an unbiased estimator of that function and has variance less than or equal to that of $\hat{\theta}$.

First, we show that $E[\hat{\theta}|T(X)]$ is unbiased. By the law of total expectation, we have:

$$E[E[\hat{\theta}|T(X)]] = E[\hat{\theta}] = \theta$$

Thus, $E[\hat{\theta}|T(X)]$ is an unbiased estimator of the function of θ .

Next, we show that the variance of $E[\hat{\theta}|T(X)]$ is less than or equal to the variance of $\hat{\theta}$. By the law of total variance, we have:

$$Var(\hat{\theta}) = E[Var(\hat{\theta}|T(X))] + Var(E[\hat{\theta}|T(X)])$$

Since $Var(\hat{\theta}|T(X)) \geq 0$, we have:

$$Var(E[\hat{\theta}|T(X)]) \leq Var(\hat{\theta})$$

Thus, the variance of $E[\hat{\theta}|T(X)]$ is less than or equal to the variance of $\hat{\theta}$.

Therefore, we have shown that $E[\hat{\theta}|T(X)]$ is an unbiased estimator of the function of θ and has variance less than or equal to that of $\hat{\theta}$. This completes the proof of the Rao-Blackwell theorem. □

Example. $X_1, \dots, X_n \sim N(\theta, \theta^2)$ with $\theta \in \mathbb{R}$ Know that $\bar{X}, s^2$ are minimal sufficient statistics for θ .

But it is not complete since $E[s^2] = \theta^2$ and $E[\bar{X}] = \theta^2/n + \theta^2$.

Example.

Remark (Evaluate Estimations). Information: One Parameter Case: Refualr paramteric Model:

$$f = \{f_\theta : \theta \in \Theta\}$$

Under Regularity Conditions, the Fisher Information is defined as:

$$I(\theta) = E \left[\left(\frac{\partial}{\partial \theta} \log f_\theta(X) \right)^2 \right] = -E \left[\frac{\partial^2}{\partial \theta^2} \log f_\theta(X) \right]$$

Theorem 7 (Cramer-Rao Lower Bound). *Let $X_1, \dots, X_n$ be a random sample from a distribution with density or probability mass function $f(x; \theta)$, and let $\hat{\theta}$ be an unbiased estimator of a function of θ . Then the variance of $\hat{\theta}$ is bounded below by the Cramer-Rao lower bound:*

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

where $I(\theta)$ is the Fisher information for θ .

Remark. If your T is complete sufficient statistic, then the UMVUE is unique and is given by $E[\hat{\theta}|T]$ and achieves the Cramer-Rao lower bound.

Theorem 8 (5.1). *The information inequality is an equality if $H(X) = T(X)$*